

Using Proactive Computing for Real-time Feedback in High-fidelity Medical Simulation

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Abstract—Using simulation for medical education improves clinical practice and creates a safe environment for students to develop skills. High-fidelity patient simulators, as one kind of simulation tool, use advanced life-like manikins. A high-fidelity simulation session includes pre-briefing, simulation scenarios, and debriefing. While pre-briefing and debriefing are widely recognized, the reflection-in-action during simulation is often overlooked despite its importance. Reflection-in-action is the ability to reflect on the student and adapt during the simulation. One major issue regarding reflection-in-action is the limited number of instructors at the university compared to the large number of students. In addition, the intensive time required for the simulations further constrains the capacity of the instructors. To address this critical gap, we propose using a proactive computing approach to provide real-time support for decisions in high-fidelity medical simulations. By implementing the proactive system and providing immediate adaptive feedback to students based on sensor data, we hope to assist teachers and improve learning outcomes.

Index Terms—High-fidelity Simulation, Proactive Computing, Medical Education, Decision Support System

I. INTRODUCTION

Learning methods and pedagogy are evolving in clinical education, with a strong emphasis on integrating technology [1]. Educational technology has been integrated into various aspects of medical education, and its use should correspond to educational needs to optimize learning outcomes [2]. Educators in health professions are at the forefront of this change and incorporate technology into their teaching practices to provide relevant, timely content in a dynamic environment [1]. "Simulation is the result of major advancements in both technology and educational theory" [3]. Using simulation for medical education can significantly improve clinical practice. It creates a safe and controlled environment where problem-based learning is promoted, and students can practice skills [4]. High-fidelity patient simulators, as one kind of simulation

tool, use advanced life-like manikins and computer programs based on complex human physiology and pharmacology mathematical models to create a dynamic and realistic simulation system [5]. A high-fidelity simulation session includes: pre-briefing, simulation scenario, and debriefing [6].

A recent study [7] discussed simulation-based education and emphasized the importance of reflective practices. Pre-briefing prepares learners by promoting reflection before the simulation. Debriefing is the critical step where students reflect on their actions post-simulation to enhance learning outcomes. However, reflection-in-action is the ability to reflect and adapt during simulation, which is vital for developing critical thinking and adaptability. The study emphasizes the significance of reflective practices, noting that while pre-briefing and debriefing are widely recognized, reflection-in-action is often overlooked despite its importance [8] [7]. In a literature review, Nielsen et al. (2013) investigated anxiety levels during the simulation, formulated recommendations to moderate anxiety, and summarized interventions into two categories: Recommendations for Faculty and Student Suggestions. One of the faculty recommendations is to "use prompts and cues as appropriate," highlighting the importance of providing students with appropriate and timely guidance during simulation exercises [9]. A 2017 study highlighted the critical role of expert instructors in simulator training in the industry, as they provide feedback and guidance, manage training scenarios, and facilitate reflection. However, the dependence on instructors can be a limitation due to their limited availability. The article proposed an intelligent tutoring system (ITS) to automate feedback in simulator training [10]. A study has used deep learning algorithms to automate feedback for suture with intracorporeal knot exercise in the fundamentals of a laparoscopic surgery simulator. Researchers developed a system that provides performance metrics, empowering surgi-

cal residents to practice independently. This system addresses the main problem of the lack of feedback during training in the absence of an expert, reducing the risk of learning bad techniques that are difficult to correct later and enabling independent practice [11].

Based on the literature we investigated above, four aspects of reflection during simulation are highlighted:

- Instructors' limitations
- Using different methods and technologies to improve reflection-in-action
- Enabling trainees to perform actions independently by receiving guidance
- Providing timely and appropriate guidance

In our previous work at the *University of Luxembourg*, focusing on improving high-fidelity simulation using technology, we have worked on various phases, including **requirement analysis** and **scenario development** [12], **pre-briefing** [13], and **debriefing** [14]. Now, a significant issue is the limited number of instructors in the university compared to the large number of students. Additionally, the intensive time required for simulations further constrains instructors' capacity.

In this work, we propose a solution that focuses on reflection-in-action and provides feedback to students during the simulation sessions. Our goal is to help teachers manage sessions effectively and provide students with immediate, clear, and adaptive guidance based on their actions. To achieve this, we have designed **proactive software to provide real-time feedback** to students during simulation by analyzing data from simulator sensors.

II. METHOD

This study uses a high-fidelity adult patient simulator developed by MedVision Group¹, designed specifically for simulation-based training in acute medical emergencies. All our research up to now has focused on a basic life support scenario (BLS) (Figure 1), a critical and guideline-based medical procedure. In collaboration with the medical department and third-year medical students, our objective is to enhance the technology and various aspects of simulation sessions using a high-fidelity manikin.

A. Automated Scenario

Instructors can use automated (pre-programmed) scenarios or manual (run-on-the-fly) scenarios. In automated scenarios, everything is pre-programmed, and the state of the manikin changes automatically based on predefined parameters while students interact with it. Manual scenarios require constant instructor control, with state changes determined by the instructor based on their real-time monitoring of the student's actions [15]. In our research, we implement and use automated scenarios, while universities typically prefer manual scenarios. Empirically, implementing automated scenarios due to limitations in current high-fidelity manikin systems is complicated, and instructors choose to use manual scenarios. However, as



Fig. 1. Implementation of the BLS Scenario in the Manikin System

we tested, automated scenarios can be implemented well for simple guideline-based scenarios, such as BLS.

Figure 1 represents the automated BLS scenario that we implemented in the high-fidelity manikin system and tested and used multiple times in our previous research. The scenario begins with a patient in a critical state exhibiting tachycardia, tachypnea, and hypotension. Over time, the patient deteriorates into ventricular fibrillation (V-Fib) and cardiac arrest (state 1). The simulation prompts the responder to assess breathing, check for pulses, and initiate airway management (State 2). Following airway management, the scenario progresses to initiating cardiopulmonary resuscitation (CPR) and defibrillation (state 3). After continued CPR cycles (State 4), the patient recovers automatic circulation, stabilizing with improved vitals (State 5). The process is trigger-driven and time-dependent.

B. Proposed Idea

The idea focuses on *Reflection-in-action* [8] during simulation sessions and to assist the instructor with immediate feedback to students. Typically, in a simulation, the instructor manages the session, guides the students, and monitors their actions. Due to time constraints and the multitasking nature of this role, managing the session is complicated. Our solution is to implement a system that complements the instructor's *reflection-in-action* by using automated scenarios and analyzing student actions in real-time. This system captures data on the fly from the manikin's sensors, provides recommendations, and helps to guide students to correct their actions. Therefore, the system not only supports instructors in better managing the simulation and freeing them to focus on other tasks but also enhances *reflection-in-action* by delivering immediate feedback to students during the simulation.

C. Gathering Requirements

To gather the system's requirements, in December 2024, we conducted a BLS simulation session with third-year students in the bachelor's program in medicine, guided by a multidisciplinary team that included an anesthesiologist, a nurse specializing in simulation education, an e-learning expert, and a medical informatics doctoral researcher. During the simulation, we documented the instructors' recommendations

¹<https://www.medvisiongroup.com/leonardo.html>

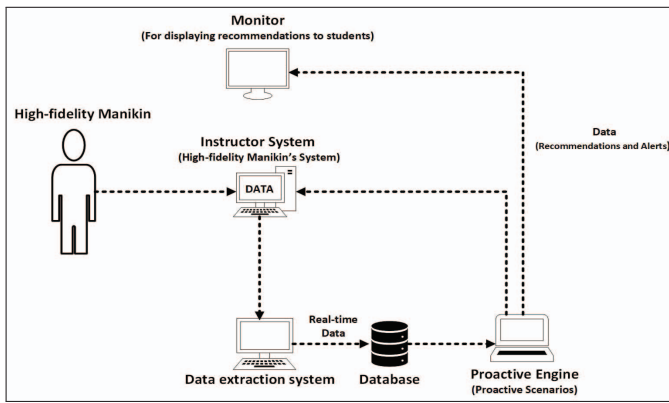


Fig. 2. Proactive system for real-time feedback to students

to the students and collected students' feedback through a questionnaire to better understand their needs.

D. Proactive System

David Tennenhouse in 2000 addressed the question of how to transition from human-centered to human-supervised computing through the concept of proactive computing. Proactive computing shifts from human-centered interaction to autonomous systems that proactively monitor and respond to their environments using sensors, and allows users to be in supervisory roles. It emphasizes real-time computing to improve efficiency and decision-making [16].

So, based on the idea of proactive computing, we are developing a proactive system, as illustrated in Figure 2, to improve reflection-in-action by providing real-time feedback to students during scenarios. The proactive software collects real-time data from the high-fidelity manikin's system via the sensors and the instructor system. It processes through the proactive engine to generate recommendations and alerts based on pre-defined scenarios and displays them on a monitor.

III. PRELIMINARY IMPLEMENTATION AND RESULTS

A. Simulation Session

A simulation session was conducted with 31 third-year medical students. The objective was for them to work as a team to perform resuscitation on a high-fidelity manikin. The students were divided into 12 groups, each consisting of 2–3 students. During the simulation session, we monitored the teacher's guidance for the students and requested the teacher to document all the guidance he offered. At the end of the session, we summarized this information in Table I. Additionally, we noted that the teacher occasionally provided cues to students when performing the actions correctly to help them remain on the right track.

On the other hand, we prepared an online open-ended questionnaire (Table II) and asked students to complete it after the session. The questionnaire included various questions addressing different phases of the simulation. In this paper, we focus on questions 4 and 5, which are related during the simulation. Our aim was to assess students' needs for guidance

during the BLS simulation session. Out of 31 students, 20 responded to the questions. We analyzed the relevant responses, excluding those unrelated or irrelevant to the questions.

Several respondents to question 4 (applying their knowledge and skills during simulation) believed in successfully applying their knowledge during the BLS simulation while highlighting significant improvement areas. The challenges were difficulties with equipment handling, unfamiliarity with processes, and stress from being observed. Many noted gaps in practical skills due to limited experience and confusion about specific procedural elements, such as asystole management or bed adjustments. However, the one that was interesting for us to see how we can make a change with a proactive system was that respondents expressed issues like being "watched" during the simulation (e.g., "I don't like being watched"). So, it came to our mind how it would be if they felt that the instructor was not watching them and instead a system was guiding them. However, it was unclear what the meaning of being watched was, by the camera or the teacher.

Question 5 (the guidance they like to receive during the simulation) was the most important to understand what the students wanted to receive as guidance during the simulation. The relevant answers were as follows: *"I think in that situation it would have been more useful if someone told us that we should focus only on chest compression and ventilation + defibrillation."* *"A person who has the role of a narrator: who says, for example: 'Ok now, Leonardo is no longer breathing, what do you do now?'"* *"During the simulation, I would like to get direct feedback about my actions so that I can improve during the simulation."* *"Answers if questions are asked (Example: are we supposed to intubate?)."* *"Be a little more guided by the microphone."*

TABLE I
ERROR POINTS HIGHLIGHTED BY THE TEACHER DURING
SIMULATION FOR STUDENT GUIDANCE

Error Type	Frequency	Notes
Inappropriate compression depth/rate	7 groups	Many performed compressions too shallow or too slow.
No defibrillation / late defibrillation	5 groups	Delays in shocking the patient.
No pulse/breathing check	5 groups	Skipped the initial assessment.
No ventilation	3 groups	Forgotten or done incorrectly.
Time lost between interventions	3 groups	Lack of coordination slowed response.
No immediate chest compressions	2 groups	Time lost before starting CPR.
Incorrect electrode placement	2 groups	Affected defibrillation effectiveness.
Leadership and communication issues	2 groups	Some teams were well-structured, others not.
No compressions during defibrillator charging	1 group	Paused compressions while charging the defibrillator, leading to interruptions.

B. Educational Psychology Consultation

To design the system, we consulted with an Educational Psychology specialist. The agreed idea for testing involves

TABLE II
SIMULATION FEEDBACK QUESTIONNAIRE

Number	Question
1	Before the simulation, did you review the pre-briefing materials in Moodle, including explanations, videos, and scenario/mannequin familiarization?
2	Have you reviewed the feedback (PDF) on your performance after the simulation?
3	Did the pre-briefing materials in Moodle prepare you for the simulation?
4	Did you effectively apply your knowledge and skills to do the BLS scenario during the simulation?
5	What kind of guidance would you prefer, and how would you like to receive it during the simulation to help you act better?
6	Was the debriefing, which included the feedback (PDF), helpful in identifying your strengths and areas for improvement?

using a monitor with a black background to display recommendations in a simple format. However, the specialist noted that this approach might divide students' attention and distract them while they are working with the manikin. It should be tested to determine its effectiveness. Another possibility is to supplement the monitor display with audio cues, similar to what an instructor might provide during a simulation, to further minimize potential distractions. The specialist emphasized that students may feel more comfortable receiving a system's guidance than feeling judged and supervised by a teacher. Furthermore, it is important to not only focus on correcting wrong actions but also provide feedback when students are performing well. This helps them to get in the right direction and keep it. Also, guidance statements should be phrased in a positive way.

C. Proactive System Primary Algorithm

A proactive system is a technology-driven framework designed to anticipate user needs, monitor conditions in real-time, and autonomously initiate actions or provide feedback to address potential issues before they arise and improve user experience [17]. Based on our observations and investigations, we designed a primary algorithm for proactive software that provides real-time guidance and feedback for the BLS scenario. It continuously monitors user actions and parameters using data received from high-fidelity manikin sensors to check adherence to standards and improve performance through immediate feedback, as outlined below (for following the steps, refer to Figure 1) :

Recognizing the Initial State: The system begins by observing the initial conditions.

Time-Based Trigger Monitoring: The algorithm tracks specific time intervals and verifies task completion:

- **Before 40 seconds:** If trigger #1 (e.g., initial checks) is not completed, the system prompts the user with:
"Check breathing and central pulse."
- **Between 40 and 70 seconds:** If trigger #2 (e.g., airway opening) is not done, the system announces:
"Open airways. Do not start CPR before opening airways."

- **After 130 seconds:** If trigger #3 (e.g., AED usage) remains unaddressed, the system advises:

"Use the AED as soon as possible."

CPR Cycles Performance Monitoring: The software evaluates multiple task parameters in real-time, such as:

- Number and depth of chest compressions.
- Rate of compressions and ventilation volume.
- Time intervals between cycles.
- Ventilation quality.

Feedback Mechanism: Based on the observed deviations, the system announces corrective actions:

• Compression Issues:

- *"Push harder; if tired, switch with your colleague."*
- *"Do the compressions slower."*
- *"Do the compressions faster."*
- *"Keep chest compressions without interruptions."*
- *"Keep chest compression number in the standard range."*

• Ventilation Issues:

- *"Adjust ventilation volume."*
- *"Provide breaths."*

D. Proactive Scenarios

Based on all the results we presented above, we have decided to create 10 proactive scenarios (Table III) for the BLS scenario, ensuring that they align with the key points highlighted by the instructor. Each scenario will include concise messages to guide students clearly and efficiently. Proactive scenarios are implemented in proactive software, which is specifically designed based on medical scenarios (BLS) within the manikin system.

TABLE III
PROACTIVE SCENARIOS

Scenario	Focus Area	Alarm/Feedback Messages
Scenario 1	Checking breathing and central pulse	Check breathing and Central pulse
Scenario 2	Opening the airway (head tilt, chin lift)	Open the airways carefully
Scenario 3	Starting chest compressions	Start Chest compression
Scenario 4	Using Automated External Defibrillator (AED)	Use the AED as soon as possible
Scenario 5	Ensuring correct number of compressions	Keep chest compression number in the standard range
Scenario 6	Maintaining appropriate compression depth	Push harder; if tired, switch with your colleague — Push less, You're pushing too much
Scenario 7	Monitoring chest compression rate	Do the compressions faster (if <100/min) — Do the compressions slower (if >120/min)
Scenario 8	Delivering proper ventilation	Give breaths
Scenario 9	Ensuring ventilation volume is within range	Ventilation volume is low — Ventilation volume is high
Scenario 10	Avoiding unnecessary pauses in compressions	Keep chest compression without interruptions



Fig. 3. The mock-up of the proactive software interface (displaying corrective message for chest compression depth).

E. Alarms and Displaying Messages

To display messages, we decided to design an interface to clearly show corrective statements that will appear in red text on a black background, accompanied by an audible alert. Positive or non-corrective feedback will be displayed in green text. What we have now accomplished is we tested a sensor related to "depth of the chest compression". The proactive software can capture data from the sensor, and if the depth of chest compression is outside the range specified in the 2021 European Resuscitation Council Guidelines [18] ("Compress to a depth of at least 5 cm but not more than 6 cm"), the system immediately processes the data and sends a recommendation. For example, if the depth of chest compression given to the manikin is less than 5 cm, the proactive system immediately sends a message like the one in Figure 3, along with an audible prompt: "Push harder."

F. Limitations

Three limitations related to the manikin system can affect the process. First, defining an automated scenario is challenging because the system features are limited and if medical scenarios are complex, not always step by step, and have multiple paths, implementing automated scenarios becomes nearly impossible. Secondly, sensors are not always reliable and can sometimes send false data and affect real-time feedback. Finally, data collection required the addition of new hardware and software components to the manikin system, ensuring that these components were seamlessly integrated with the existing infrastructure.

IV. CONCLUSION

The paper discusses our idea of developing proactive software for reflection-in-action. This system could potentially improve learning outcomes by providing immediate decision support through real-time analysis of data from multiple sensors and addressing issues related to the limited number of instructors and the intensive time required for simulations. The purpose is to help teachers manage sessions better and provide students with immediate, clear, and prompt guidance based on their actions. In the future, we propose adapting the system to various scenarios and expanding it to address advanced life support (ALS) and more complex medical emergencies. In addition, evaluations of student and teacher satisfaction should include a controlled comparative study to validate the effectiveness of the system.

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